Lab 6: Code and Branch Coverage Analysis

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Objective

- Understand the difference between **code coverage** and **branch coverage**.
- Use coverage to **measure**, **analyse**, and **improve** test coverage.
- Interpret coverage reports and identify **untested paths** in code.

Learning Outcomes

- **Understand Code Coverage Concepts**: You will differentiate between line coverage, branch coverage, path coverage, and function coverage to measure test completeness.
- **Apply Coverage Analysis Tools**: You will use Python's coverage library to generate reports, interpret coverage metrics, and identify untested code paths.
- **Design Comprehensive Test Cases**: You will write targeted test cases to achieve minimum 90% to 100% coverage by addressing edge cases and decision branches.

Introduction

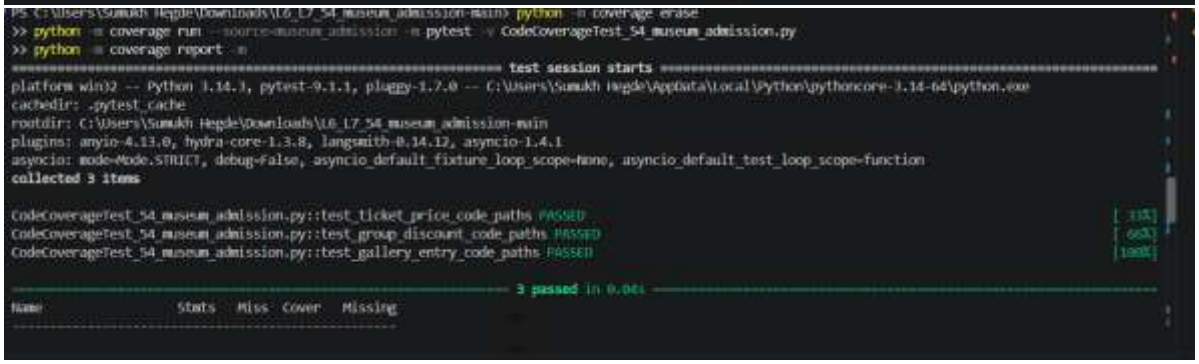
What is Code Coverage?

Code coverage measures how much of your source code is **executed during testing**.



```
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main> run recent commands
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main> python -m pip install pytest coverage
Requirement already satisfied: pytest in C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (9.1.1)
Requirement already satisfied: coverage in C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (7.17.0)
Requirement already satisfied: colorama>=0.4 in C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pytest) (0.4.6)
Requirement already satisfied: iniconfig>=1.0.1 in C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pytest) (2.3.1)
Requirement already satisfied: packaging>=22 in C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pytest) (26.0)
Requirement already satisfied: pluggy<2,>=1.5 in C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pytest) (1.7.0)
Requirement already satisfied: pygments>=2.7.2 in C:\Users\Sumukh Hegde\AppData\Roaming\Python\Python314\site-packages (from pytest) (2.20.0)
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main> python -m pytest -v
============================= test session starts ==============================
platform win32 -- Python 3.14.3, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main
plugins: anyio-4.13.0, hydra-core-1.3.8, langsmith-0.14.12, asyncio-1.4.1
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 2 items / 1 error
```

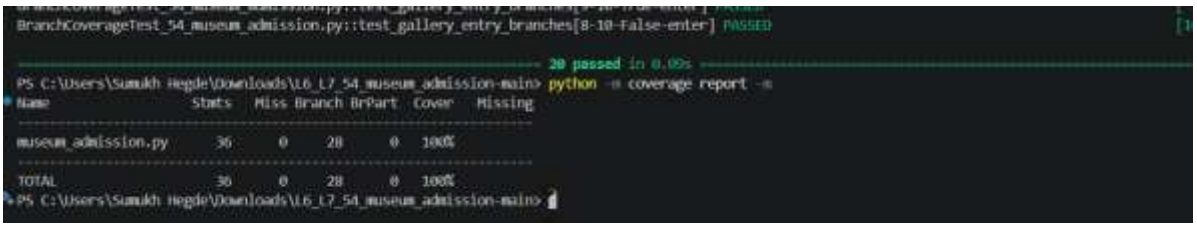
```
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main\LAB6> python -m coverage erase
>> python -m coverage run --source=museum_admission -m pytest -q test_museum_admission.py
>> python -m coverage report -m
..
2 passed in 0.01s
Name                 Stmts   Miss  Cover   Missing
--------------------------------------------------
museum_admission.py      36     19    47%   6, 8, 10, 13, 15, 17-20, 26, 28-30, 32, 34, 39-43
--------------------------------------------------
TOTAL                    36     19    47%
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main\LAB6> python -m coverage erase
>> python -m coverage run --branch --source=museum_admission -m pytest -q test_museum_admission.py
>> python -m coverage report -m
..
2 passed in 0.01s
Name                 Stmts   Miss Branch BrPart  Cover   Missing
```



```
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main> python -m coverage erase
>> python -m coverage run --source=museum_admission -m pytest -v CodeCoverageTest_54_museum_admission.py
>> python -m coverage report -m
============================= test session starts ==============================
platform win32 -- Python 3.14.3, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\Sumukh Hegde\AppData\Local\Python\pythoncore-3.14-64\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main
plugins: anyio-4.13.0, hydra-core-1.3.8, langsmith-0.14.12, asyncio-1.4.1
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 3 items

CodeCoverageTest_54_museum_admission.py::test_ticket_price_code_paths PASSED                                                [ 33%]
CodeCoverageTest_54_museum_admission.py::test_group_discount_code_paths PASSED                                              [ 66%]
CodeCoverageTest_54_museum_admission.py::test_gallery_entry_code_paths PASSED                                               [100%]

============================== 3 passed in 0.04s ===============================
Name                Stmts   Miss  Cover   Missing
```



```
BranchCoverageTest_54_museum_admission.py::test_gallery_entry_branches[8-10-False-enter] PASSED                            [1

============================== 20 passed in 0.05s ==============================
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main> python -m coverage report -m
Name                 Stmts   Miss Branch BrPart  Cover   Missing
--------------------------------------------------------------------
museum_admission.py      36      0     28      0   100%
--------------------------------------------------------------------
TOTAL                    36      0     28      0   100%
PS C:\Users\Sumukh Hegde\Downloads\L6_L7_54_museum_admission-main>
```